

Lumped-Parameter Modeling and Calibration of Hexapole Electromagnetic Actuators for Magnetic Flux Generation

Current Base Model

$$b(p) = \underbrace{\left(\frac{6}{5}k_{11}\right) \left(\frac{N_c}{4\pi\widehat{\ell}^2 R_a}\right)}_{B\widehat{g}_I} S_i \overline{K}_I I = S_i \underbrace{\left(B\widehat{g}_I \overline{K}_I I\right)}_G$$

Voltage Base Model

$$b(p) = S_i \cdot D \cdot V = S_i \cdot \left[\underbrace{\left(\frac{6d_{11}}{5}\right) \overline{D} V}_{B\widehat{g}_V} \right] = S_i \underbrace{\left(B\widehat{g}_V \overline{D} V\right)}_G$$

Symbol	Unit
R_a	$\frac{A}{Wb}$
I	A
$B\widehat{g}_I$	$\frac{mT}{A}$
$B\widehat{g}_V$	$\frac{mT}{mV}$
V	mV
$\widehat{\ell}$	μm
$b(p)$	mT
$\left(\frac{3V}{\mu_0}\right) \left(\frac{\mu - \mu_0}{\mu + 2\mu_0}\right) \triangleq M\widehat{g}_B$	$\frac{\mathbf{m}}{mT}$
$\left(\frac{M\widehat{g}_B}{2\widehat{\ell}}\right) (B\widehat{g}_I)^2 \triangleq F\widehat{g}_I$	$\frac{pN}{A^2}$
$\left(\frac{M\widehat{g}_B}{2\widehat{\ell}}\right) (B\widehat{g}_V)^2 \triangleq F\widehat{g}_V$	$\frac{pN}{mV^2}$
$\mathbf{m}(p) \triangleq M\widehat{g}_B \cdot b(p)$	$A \cdot \mu m^2$
$\begin{cases} k_m = \frac{\mu_0}{4\pi} = 10^{-7} \\ \mu_0 = 4\pi \cdot 10^{-7} = 0.0013 \cdot 10^{-3} \end{cases}$	$\frac{pN}{A^2}$